Lead Scoring Assignment – Subjective Questions

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Model used for every answer: logistic regression with 23 variables (RFE → p-value < 0.05 → VIF < 5). On unseen test data: AUC 0.976, accuracy 93.5%, sensitivity 92.9%, specificity 93.9%; lead score = probability × 100; hot lead = score above 35 (cut-off 0.35). **How to read the numbers:** a coefficient is the change in log-odds versus the reference level (e.g. "Not Specified", "Google"); odds ratio = e^{coef}; time spent is standardised, so its coefficient is per one standard deviation.

Q1

"Which are the top three variables in your model which contribute most towards the probability of a lead getting converted?"

Rank	Variable	Coefficient	Odds ratio	What it means
1	Tags: Lost to EINS	+5.32	x 205	Sales tag for a lead that already decided (97.7% of them converted)
2	Tags: Closed by Horizzon	+5.07	x 158	Sales tag for a closed deal (99.4% converted)
3	Lead Source: Welingak Website	+3.18	x 24	Leads from this partner site convert at 98.6% versus 40% for Google

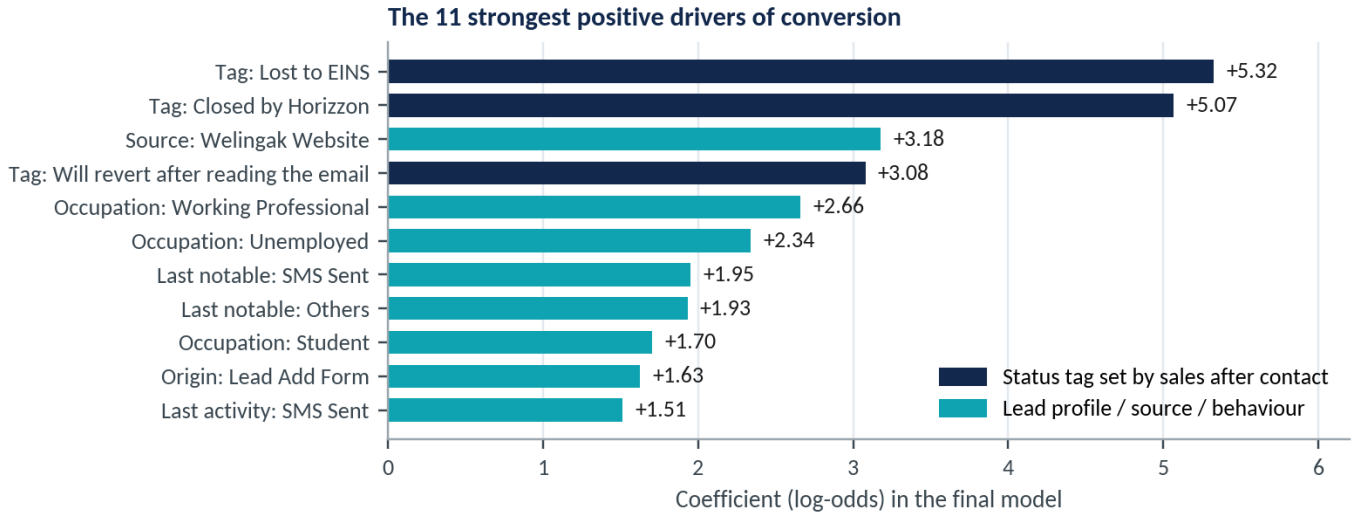
Why these three: I ranked all 23 variables by coefficient, because the numeric and dummy variables are all on the log-odds scale. The first two are statuses that the sales team records once a lead has already decided, so they *confirm* the outcome rather than cause it – they are the model's strongest signals, not levers. The most important **behavioural** variable is **Total Time Spent on Website** (+1.08 per standard deviation, odds x 2.9): converted leads spend on average 739 time units on the site against 330 for the rest.

Q2

"What are the top 3 categorical/dummy variables in the model which should be focused the most on to increase the probability of lead conversion?"

Rank	Dummy variable	Coefficient	Odds ratio	Where to focus
1	Lead Source: Welingak Website	+3.18	x 24	Grow this partner/referral channel; give its leads priority calls
2	Occupation: Working Professional	+2.66	x 14	Aim ads and course messaging at working professionals
3	Lead Origin: Lead Add Form	+1.63	x 5	Push visitors to fill a form; these leads are already interested

Why these three: the Tags dummies are left out because they are set by sales after contact, so the company cannot "focus on" them. Among the dummies the business can influence, these have the largest effect. Next best is **Last Activity: SMS Sent** (+1.51, x 4.5), which is fully in the sales team's control. "Unemployed" also has a large coefficient (+2.34), but that mostly reflects the contrast with leads who left the occupation blank (only 14% of them convert), so it is not a segment to chase. **Practical advice:** also make the occupation field mandatory in the form, since a blank answer is itself a warning sign.



Bottom line for Q1–Q2: engagement (time on site), the source and origin of the lead (partner site, lead form) and the lead's occupation are the levers the company controls; the sales tags are the strongest *confirmation* that a lead is about to buy or has been lost.

Q3

"X Education has a period of 2 months every year during which they hire some interns. The sales team has around 10 interns allotted to them. So, during this phase, they wish to make the lead conversion more aggressive. So, they want almost all the potential leads (i.e., the customers who have been predicted as 1 by the model) to be converted and hence, want to make phone calls to as much of such people as possible. Suggest a good strategy they should employ at this stage."

Strategy: lower the cut-off from 0.35 to about 0.15 (score of 15 or more) so that the model marks more leads as "potential", and give the extra leads to the interns.

Why: a lower cut-off raises **sensitivity** – the share of real buyers that get a call. On the test set, the team then calls 1,219 leads instead of 1,096 (only 123 extra calls) and reaches **96.1%** of all buyers (up from 92.9%). The conversion rate of the calls stays at **84%**, above the CEO's 80% target. 0.15 is the natural floor: at 0.10 the rate drops to about 78%.

How to run it: (1) experienced reps take scores 81–100 first (95% convert); (2) interns take the 16–80 band – 297 leads, of which the 16–35 lukewarm group converts at 28% and the 36–80 warm group at 66%, still 10–25 times better than the 2.7% of the cold group; (3) review the conversion rate of calls weekly and move the cut-off back up if it approaches 80%.

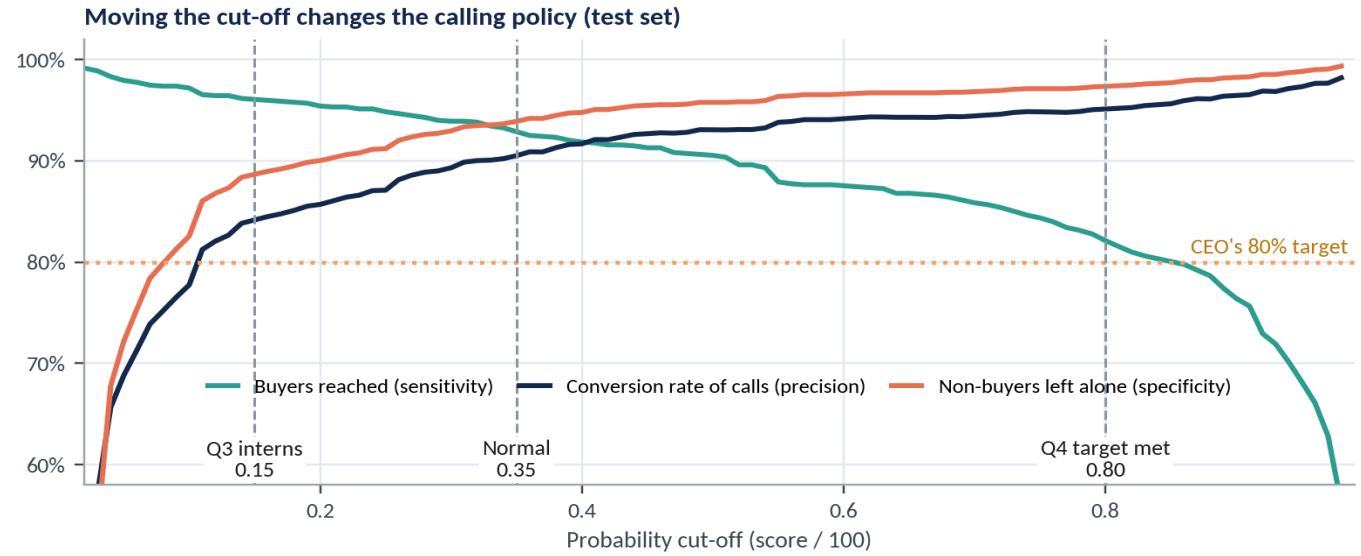
Q4

"Similarly, at times, the company reaches its target for a quarter before the deadline. During this time, the company wants the sales team to focus on some new work as well. So, during this time, the company's aim is to not make phone calls unless it's extremely necessary, i.e., they want to minimize the rate of useless phone calls. Suggest a strategy they should employ at this stage."

Strategy: raise the cut-off to about 0.80 (score of 80 or more) and phone only those leads; move everybody else to automated e-mail/SMS nurturing.

Why: a high cut-off raises **specificity and precision**, so almost every call is to a buyer. On the test set only 33% of leads are called, **95% of those calls convert**, and useless calls fall from 104 to 45 (a 57% cut). The trade-off is that about 18% of buyers are not phoned this quarter, which is acceptable because the target is already met. The leads scoring 36–80 stay in the e-mail/SMS flow and are picked up again in the next cycle; the freed sales time goes to the new work.

How to run it: (1) the reps phone only the 922 leads scoring 80 or more, highest score first; (2) leads scoring 36–80 receive an automated e-mail/SMS sequence, skipping those who ticked "Do Not Email"; (3) leads below 35 are not contacted at all (only 2.7% of them convert); (4) when the new work ends, go back to the 0.35 cut-off.



Calling policies at a glance (test set, 2,772 leads, 1,068 real buyers)

Policy	Cut-off	Leads called	Conversion rate of calls	Buyers reached	Useless calls
Call everyone (no model)	–	2,772 (100%)	38.5%	100.0%	1,704
Intern season (Q3)	0.15	1,219 (44%)	84.2%	96.1%	193
Normal operation	0.35	1,096 (40%)	90.5%	92.9%	104
Target reached (Q4)	0.80	922 (33%)	95.1%	82.1%	45

Assumptions: Tags are recorded by the sales team during nurturing, so the model is a nurturing-stage scorer (without Tags, AUC falls to 0.88); cut-offs were chosen on the training data and figures come from the 30% test set; the CEO's 80% target is read as the conversion rate of the leads that are actually called.